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Parametric Control of Prosodic Variables by Symbolic Input in TTS Synthesis

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ABSTRACT This chapter outlines the prosodic module of the RULSYS/INFOVOX TTS system for German, which represents the implementation of KIM (the Kiel intonation model). The first section sets out the categories of this model, and the second deals with their symbolization in input strings to the model and to the synthesis system for the control of parametric values.

37.1 KIM – The Kiel Intonation Model

37.1.1 Overview

The prosody model that we have been developing at Kiel for German and that has been implemented in the RULSYS/INFOVOX text-to-speech system incorporates the following domains (for further details see [Koh91]):

1. lexical stress—three levels: unstressed, secondary stress in compounds, and primary stress
2. sentence stress—four levels: reinforced, accented, partially deaccented, and completely deaccented
3. intonation:
    a. categories of pitch peaks and valleys as well as their combinations at each sentence stress position
    b. types of pitch category concatenation
4. synchronization of pitch peaks and valleys with stressed syllables—three steps: early, medial, late
5. prosodic boundaries (degrees of cohesion)—three variables: pause duration, phrase-final segmental lengthening, scaling of F_0 end points
6. overall speech rate between the utterance beginning and successive prosodic boundaries—four degrees: slow, medium, reduced, fast

7. downstep of successive pitch peaks/valleys and pitch reset

A system of prosodic distinctive features is used to specify the abstract symbolic, phonological categories in these domains. They are attributed to phonological units, which are either segmental (vowels and consonants) or nonsegmental (morphological and phrase boundaries). Attached to vowels is the fundamental distinction within the German prosodic system, namely, stress and intonation.

37.1.2 Stress

Lexical Stress

Within stress we have to differentiate between lexical and sentence stress. At the abstract level of phonological specifications in the lexicon, every German word has at least one vowel that has to be marked as potentially stressable, as being able to attract the feature specifications of sentence stress. Lexical stress is thus not a distinctive stress feature, it only marks a position that can, but need not, attract such a feature at the sentence level.

In noncompounded words one vowel has primary lexical stress. In compounds this also applies to at least one component word; the remaining compound word elements each have one vowel with secondary lexical stress. All vowels that are neither primary nor secondary-stressed are unstressed at the lexical level.

Sentence Stress

Sentence stress is attributed to a word as a whole; the lexical stress position in the word determines where the sentence stress manifests itself. By default a nonfunction word receives the category "accented." Deviations from this are either in the direction of emphatic reinforcement or of deaccentuation, which may be partial or complete. Function words are by default unaccented (= completely deaccented). Deviations are partially (de)accented, accented or reinforced.

37.1.3 Intonation

Pitch Categories at Sentence Stresses

All vowels of the reinforced, accented, and partially (de)accented sentence stress categories receive intonation features, which may be either valleys or peaks. Peaks may contain a unidirectional F_0 fall, classified as terminal, or rise again at the end, resulting in a (rise-) fall-rise, categorized as nonterminal. Rises in valleys and nonterminal peaks may be low, to indicate, for example, continuation, or high, for example, used in questions. All peaks and valleys may have their turning points (F_0 maximum in peaks or F_0 minimum in valleys) early or later with reference to the onset of the sentence stressed vowel, categorized as "(non-)early," and finally, for peaks, non-early may be around the stressed vowel center or toward its end, classified as "medial" or "late." This categorization of peaks captures the grouping

of late and medial versus early peaks, as it showed up in perceptual experiments with stepwise peak shift from left to right [Koh90b].

Peaks are characterized by a quick F_0 rise confined to the vicinity of a sentence-stressed syllable. This rise precedes the onset of the latter, and is usually short and narrow in range for an early peak; it extends into the first half of the stressed nucleus in the case of a medial peak. In the late peak, it starts after the stressed vowel onset and continues into the second half of the nucleus or beyond. The exact timing of the maximum peak value depends on vowel type (duration according to quantity and quality), subsequent voiced/voiceless consonants, and number of immediately following unstressed syllables. There may even be a low stretch of F_0 in the stressed vowel before the rise. After the peak maximum is reached the F_0 descends immediately, especially on subsequent unstressed syllables. For chains of peaks, see subsection 37.1.3.2 below.

Valleys, on the other hand, have a continuous rise, starting before the stressed-syllable nucleus (early) or inside it (nonearly) and extending as far as the beginning of the following sentence-stressed syllable. If there are several unstressed syllables between two sentence stresses, a valley is thus realized as a more gradual F_0 ascent compared with the much quicker rise for a late peak. The less distance there is between stressed syllables, the more difficult it becomes to distinguish between a valley + peak and a late peak + peak sequence, especially if there is no F_0 dip between the first and second stress F0 maxima, as in a hat pattern (see 37.1.3.2).

Pitch Category Concatenation

In a concatenation of pitch peaks without prosodic boundaries between them (see subsection 37.1.4.3), F_0 may fall to a low or an intermediate level and then rise again for the next peak. This fall will be effected on intervening unstressed syllables between the two peaks, reaching the lowest point, to start the next rise, in the vicinity of the following stressed syllable, depending on peak position. If there are no unstressed syllables separating the two peaks, the dip can be accommodated between all peak combinations, except for late + early/medial, where a hat pattern is created; it combines the rise of the late peak and the fall of the early peak in a two-stress sequence.

This boundary case of the absence of an F_0 descent between peaks can also be extended to concatenations with intervening unstressed syllables. In such a hat pattern, an early peak is not possible initially, and a late one is excluded noninitially. If there are more than two stresses incorporated in a hat, the noninitial and nonfinal ones are unspecified as to peak position because they have neither a rise nor a fall but are simply integrated into the downstepped sequence of peak maxima (see 37.1.4.2). In the categorization of pitch patterns they are nevertheless grouped together with peaks. If in a two-stress rise-fall it is difficult to decide whether the rise represents a valley or a late peak in a hat pattern, the latter is chosen.

When prosodic boundaries intervene, any sequencing of peaks and/or valleys is possible, but the hat pattern is then excluded because it represents a very high degree of cohesion. On the other hand, a late peak with a full F_0 descent marks a

dissociation from a following peak and will then normally be linked with a prosodic boundary, that is, final lengthening and F_0 reset afterwards.

Prehead

Unstressed syllables preceding the first sentence stress in a prosodic phrase may be either low or high: they represent different types of prehead.

37.1.4 Parametric Phonetic Control

The acoustic manifestations of the distinctive prosodic categories vary according to segmental and prosodic context, depending on the temporal alignment of peaks and valleys—defined by a small number of significant F_0 points—with different syllable types and sequences, on downstepping, speech rate, prosodic boundaries, and, finally, articulation-induced microprosody.

Temporal Alignment of Peaks and Valleys

Taking the default, medial peak as a reference, two significant F_0 points are defined. The first one, TF_0, is positioned at the beginning of the syllable containing the sentence-stressed vowel; the second, T2F_0, near the vowel center, the exact timing after voiced vowel onset depending on vowel quantity, vowel height, number of following unstressed syllables, and position in the utterance. The calculation of the time point T2F_0 after vowel onset is carried out on the basis of the segmental duration rules for German. They have adopted the principle proposed by Klatt [Kla79] for the rule synthesis of English (see also [Koh88]).

T2F_0 for medial peaks is derived from the basic vowel-type–related duration. It is essentially the intrinsic vowel duration that determines the point in time after stressed-vowel onset that it is positioned. But this has to be adjusted in the case of aspiration. On the one hand, aspiration lengthens the total vowel duration, compared with vowels in nonaspirated contexts, but this increase is not as large as the total aspiration phase; on the other hand, it shortens the stop closure duration compared with unaspirated cases, but again not by the total amount. So the larger part of the aspiration (three-quarters, according to comparative data analysis in the model construction [Koh91]) should be added to the vowel, but the rest attached to the plosive, and the F_0 peak placement has to take this ambivalence into account. That is, T2F_0 is shifted to the right by the period of aspiration time added to the vowel duration.

Sentence-final medial peaks receive a third F_0 point, T3F_0, which has been ascertained by data analysis and perceptual testing [Koh91] to occur 150 ms after the peak maximum in a medium speech rate (see 37.1.4.4, as well as item (b) in section 37.1.1 and item (b′) in section 37.2.1). In all nonfinal cases, the peak maximum of one sentence stress connects with the left-base point of the next sentence stress. As the absolute peak position is not affected by vowel duration modifications due to voiced/voiceless context, number of syllables in the word, sentence position, and so on, its relative position changes with vowel shortening

or lengthening, moving closer toward or further away from the end. This way the microprosodic F_0 truncation before voiceless obstruents is automatically built into the rules.

Thus, in utterance-final short-vowel monosyllables ending in voiceless consonants, an F_0 peak fall is truncated, and has to be so to create the same perceptual peak pattern as in other, nontruncating contexts. The listener obviously takes the underlying constancy in absolute F_0 peak positioning within the same vowel type, and the difference across different vowel types, into account, disregarding contextual adjustments. He can always calculate the final F_0 point that should have been reached if the F_0 contour had not been curtailed, from the F_0 decline per unit of time up to the cut-off, and he can then compare this value with the likely low end of the speaker's speech voice range (about 60–80 Hz in a male voice, an octave higher in a female one). If the comparison is within a narrow margin the fall is terminal, otherwise it is not. Since the lower end of a speaker's F_0 range is a reliable reference value, truncation of falling F_0 patterns can be uniquely restored perceptually.

That no longer applies to rising F_0. Here the end point is not calculable because there is a large margin for the end point; anything up to or even above 1.5 octaves is possible. The position of the ceiling is not fixed, in contrast to the base line. That means that the intended high value of a valley always has to be physically reached, it cannot be deduced from what precedes, and, moreover, it does not change with different positions from early to late, which is different in peaks. There is thus a fundamental difference between peaks and valleys in the fixation of their offsets.

An early peak has its maximum value at the stressed-syllable onset, TF_0 100 ms before, and T3F_0—in sentence-final position—in an area where the medial peak has its maximum. A late peak has TF_0 at the same point as a medial peak, then an additional low F_0 point T2F_0 is inserted at vowel onset, and the late summit (T3F_0) occurs 100 ms after the point where a medial peak would have its center, or at the end of the last voiced segment in a nonfinal monosyllabic word if this distance is less than 100 ms. If there is an unstressed syllable following, the summit coincides with the unstressed vowel voice onset. In utterance-final position, a fourth F_0 point, T4F_0, occurs 100 ms after the summit; in monosyllables without final voiced consonants, T3F_0 has to occur at least 30 ms before vowel offset to signal the F_0 descent to T4F_0. The time positions of these significant F_0 points were obtained by interactive perceptual evaluation using the TTS development platform (see section 37.3).

Valleys have their left and center F_0 points at the same positions as TF_0 and T2F_0 in medial peaks; the right high point is located at the end of the last voiced segment.

Downstepping

Declination, the temporally fixed decline of F_0 prominent in, for example, the Dutch and Lund models [tCC90, Gar86], is not a feature of spontaneous speech production. It has therefore been replaced by downstepping in KIM, that is, a

structurally determined pitch lowering from sentence stress to sentence stress, independent of the time that elapses between them.

Results of interactive testing make it clear that perception orient itself at structurally positioned and downstepped peaks, not at a time-based declination. For medium speech rate, the downstepping values used in the TTS implementation of KIM are 6% from peak to peak (starting from 130 Hz in a male voice), and 18% from a peak maximum to the subsequent base of the peak configuration. In valleys both the low and the high F_0 value are downstepped by 6%. Downstepping can be interrupted at any point by the reinforcement feature or by a resetting; it is stopped at a threshold value if the significant point(s) would go below it (set at 95 Hz for peak maxima in a male voice). Because downstepping is treated as automatic in the model, it is not given a phonological feature, nor is it symbolized.

Prosodic Boundaries

One of the functions of prosody is the sequential structuring of utterances and discourse, i.e. the signaling of prosodic boundaries and—at least partially—their hierarchical organization. To decode the syntagmatic chunking of messages in accordance with the speaker's intention, the listener requires signals that index degrees of cohesion or separation, respectively, between phrases, clauses, utterances, and turns. The parameters that achieve this are pause duration, phrase-final segmental lengthening, and scaling of F_0 end points at the respective boundaries. They can be controlled by parametric rules in the prosodic model upon appropriate symbolic input.

As at this stage the linguistically and phonetically relevant categorization of these boundaries is not well understood. The modeling cannot reduce the categories in this domain to the same small number as in the other areas of prosody discussed so far, but has to allow sufficient degrees of freedom for experimentation with data modeling. At each of the three parameters three degrees are therefore recognized, controlled by digit notation in the symbolic input to the model (see item (5′) in section 37.2.1). As our knowledge of prosodic boundary marking increases, the degrees of freedom can be reduced by establishing constraints between the three parameters in the signaling of the necessary and sufficient number of phonologically relevant distinctions.

Speech Rate

Speech rate changes within the same speaker not only alter the segmental durations (at varying degrees for different segment types, e.g., vowels versus consonants) but also the positioning of T2F_0 within a stressed vowel, moving it farther or less far into it in slower and faster speed, respectively (see subsection 37.1.4.1). This also implies slower or faster rises and falls, which lower or raise the perceived pitch level. The faster movements in turn mean that there comes a point when the complete F_0 excursions can no longer be performed. Since the peak and valley maxima are the essential target values controlled by the speaker, the leveling of F_0 movements in fast speech will particularly affect the low values. Finally, faster

speech also means greater effort, which may produce a higher level of activation at the vocal folds right away. The correlation of F_0 level with speech rate perception has been shown in [Koh86b, Koh86c].

The control of speech rate is also coupled with articulatory reduction and elaboration [Koh90a]. The speech rate category in KIM and in its synthesis application therefore activates whole blocks of parametric rules that deal with F_0 timing, F_0 patterning, segment durations, and segmental adjustments (coarticulation, reduction, reinforcement, elision). To start with, the model distinguishes four degrees, one of them—reduced—taking reduction phenomena (at an otherwise medium speech rate timing) into account, as one way of changing speed of articulation.

Microprosody

In the generation of intonation, KIM separates two levels:

- the defining of phonology-controlled prosodic patterns by a small number of significant F_0 points
- the output of continuous F_0 contours influenced by articulation-related modifications [Koh90b]

This dichotomy implies the assumption that the underlying F_0 peak and valley patterns develop independently and in a very concrete physical and physiological sense in speech production, and are modified microprosodically by output constraints in the vocal apparatus. In particular, the model distinguishes five areas of microprosodic adjustments to the basic significant point patterns discussed so far.

1. In close vowels with medial peaks, the summit is raised by a factor of 1.08 compared with all other vowels.
2. An interpolation (cosine) is carried out between the significant F_0 points.
3. After voiceless obstruents, the F_0 value at vowel onset is raised by an additive constant of 15 Hz, the increment trailing off to 0 toward T2F_0.
4. In voiced plosives, all F_0 values are lowered by 10 Hz; in other voiced consonants by 5 Hz.
5. F_0 is masked in voiceless stretches.

37.1.5 Linguistic Environment of the Model

KIM is integrated into a pragmatic, semantic and syntactic environment. The input into the model consists of symbolic strings in phonetic notation with additional pragmatic, semantic, and syntactic markers. The pragmatic and semantic markers trigger, for example, the pragmatically or semantically conditioned use of peak and valley types or of sentence focus. Lexical stress position can largely be derived by rule; syntactic structure rules mark deaccentuation and emphasis in word, phrase,

clause, and sentence construction. Phrasal accentuations are thus derived from the syntactic and semantic components that precede the prosodic model, and are given special symbolizations in the input strings to the model.

37.2 Symbolization of the Prosodic Categories

37.2.1 Symbolic Input to the Model

The following 7-bit ASCII characters are used to represent the prosodic categories of section 37.1.1 (using the corresponding numbering).

(1') Apostrophe or quotation mark [' "] are put in front of the primary or secondary stress vowel; vowels without a lexical stress marker are unstressed:

R'ŭck#s"icht (view to the rear) versus *R'ŭcksicht* (consideration)

(# marks the phonetically—especially prosodically—relevant word boundary in compounds.)

(2') Digits [3,2,1,0] are put in front of words that receive the reinforced, accented, partially, or completely deaccented sentence stress category, respectively, which in turn affects the manifestation of the respective lexically stressed vowel. Function words, marked by suffixed [+], have [0] as their default, nonfunction words [2]; in both cases the digit may be omitted from the symbolization:

2Max 0hat+ 0einen+ 2Brief 2geschrieben
(Max did write a letter)

2Max 0hat+ 0einen+ 2Brief 1geschrieben
(semantically unmarked rendering of Max wrote a letter)

2Max 0hat+ 0einen+ 2Brief 0geschrieben
(Max wrote a letter, not a card)

2Max 0hat+ 0einen+ 3Brief 0geschrieben .
(reinforcement of contrasted Brief in the previous example)

(3') Punctuation marks [. , ?] for pitch peaks, low and high rising valleys, and the character sequences [.,] and [.?] for fall-rises are put before prosodic boundaries (see (5')):

Ja .p: Ja ,p: Ja ?p: Ja .,p: Ja .?p:

A high prehead is symbolized by [=] after the prosodic boundary marker [p:] (see (5')).

p:= 0Wie+ 0sieht 0das+ 0bei+ 2Ihnen+ 0am+ 3Donnerstag 0aus .p:

(4') Parentheses [)(] for early and late peak positions, respectively, in sentence-stress syllables are put before the stressed word (after the sentence-stress digit); the medial peak position is regarded as the default case and remains unmarked:

Sie+ hat+ ja+ 2)gelogen
(She's been lying = summarizing, concluding statement)

Sie+ hat+ ja+ 2gelogen
(=start of a new argumentation)

Sie+ hat+ ja+ 2(gelogen
(as the preceding example but with a contradictory note)

In connection with valleys, there are only early and nonearly positions in the sentence-stressed syllable. Either one or the other category may be taken as the unmarked default case, depending on their frequency of occurrence: early for [,], nonearly for [?].

(5') The prosodic boundary (cohesion) marker [p:] is put after the word at which boundary indices occur. It is preceded by two digits, the second of which refers to pause length, the first to utterance-final lengthening. In the case of pitch peaks, there is a third boundary-related digit to the left of these two, referring to the scaling of the F_0 end point. Each of the digits may range from [0] (= absence of pause, of final lengthening or of F_0 descent) through [1] (= short pause - $\leq$ 200ms; default utterance-final lengthening; intermediate F_0 descent) to [2] (= long pause - $\geq$ 200ms; hesitation lengthening; full F0 descent):

zehn .212p: (110p:) minus+ zwei .0/100p: mal+ drei
$(10 - 2 \times 3)$
zehn .0/100p: minus+ zwei .211p: (.110p:) mal+ drei
$((10 - 2) \times 3)$

(6') The digit string associated with the phrase boundary marker [p:] is preceded by a further digit, ranging from 0 to 3 to mark four degrees of speech rate, which include degrees of reduction or elaboration: [2] refers to medium overall speed and default reduction (and may be omitted from the symbolization as an implicit default), [1] refers to the same speed but a higher degree of reduction; for [0], degrees of reduction and speed are increased, for [3] they are both decreased from [2]. The rate digit (or default) applies to the stretch of speech between the [p:] marker and its predecessor or the utterance beginning, respectively. In this modeling of speech rate, segment durations are not changed by uniform and proportionate up- or down-scaling across the whole sequence, but vowels and consonants are dealt with separately according to sets of rules including segmental reduction, assimilation, and elision.

mit+ roten gelben blauen schwarzen .0-3212p:
(with red, yellow, blue and black ones)

(7') Downstep is not indicated symbolically. Pitch reset is associated with a prosodic boundary. It is marked by [+] before the digit sequence at the preceding [p:].

mit+ roten gelben .+2110p: blauen schwarzen .2212p:

37.2.2 *Symbolic Input in the German RULSYS/INFOVOX TTS System for Parametric Control*

KIM has been implemented in the RULSYS/INFOVOX TTS for German. The Kiel development of this TTS system (for details see [CGH90, Koh91]) makes use of a very simple adaptation of 7-bit ASCII to the phonetic transcription of German:

- upper-case letters for segmental phonemes
- lower case with for allophones

1. [•] the characters listed in (1')–(7') of section 37.2.1, but with [%] replacing the compound boundary marker [#].

These phonetic symbols are either derived by rule from orthographic input or entered into the system directly, enclosed between the metacharacter #. In the latter case the input string can be either entirely phonetic or mixed orthographic/phonetic, as illustrated by the examples in the Appendix.

The TTS application of the prosodic system is based on default assumptions for certain orthographic characters that are identical with prosodically defined symbols, e.g., <?> - [?]:

- Orthographic <?> is rule-converted into phonetic [.] in question-word, but into phonetic [?] in yes-no questions.
- The deviations from default have to be specially indicated
  (a) by [.] in *Kommt sie*, for example
  (b) by [?] combined with the special marking of the otherwise default stress [2] category, in *Wie 2heißt du?* for example
  (c) by [?] combined with the special marking of the stress [2] category on the question word of a confirmatory question, which has a continuous F_0 rise throughout

In peaks the medial position is treated as default; in low valleys it is the early, in high valleys the late postion.

The period <.> has two functions in the TTS-system:

- a trigger symbol for the generation of terminal intonation patterns, that is, various degrees of falling F_0

- a command to the system (by the side of <?> and <!>) to start sentence processing

These functions cannot be dissociated from each other. This means that the input string must not contain a non–sentence-final <.> because the string following <.> will be processed separately, and, on the other hand, every sentence must be terminated by either <.> or <?> or <!> to initiate the sentence-level processing. So if a low rise, symbolized by <,>, is to occur in sentence-final position, the notation must be <,.>. This also implies that [p:] must precede punctuation marks in the TTS input strings (as against the KIM notation in section 37.2.1 (3')), and that [.] in the function of a sentence-medial falling intonation category (see (5') of section 37.2.1) cannot be input into the system as <.> but must be generated internally from [p:] if the latter stands on its own, that is, is not followed by <,>. A further necessary adaptation of the KIM symbolization system in its TTS implementation concerns the falling-rising patterns, which in KIM are mnemonically represented as [.,] and [.?]. The adjustment uses <„> and <,?> for falling + low/high rising, and sentence-final <„> is followed by <.> in the input string.

In the KIM symbolization system, intonation and prosodic boundary markings are kept strictly separate (see section 37.2.1 (3'), (5')), whereas in TTS, punctuation marks receive a (pause) duration: <,>=300ms, <. ? !>=600ms. This means that #(2)212p:.#, as default, may be reduced to <.> in the TTS input string, similarly #(2)12p:,# to <,>. And the pause length between TTS sentences can be graded in 600-ms steps by the addition of the respective number of <.> to the sentence-final punctuation mark (see Appendix, II. Texts).

The greater part of the prosodic notations in section 37.2.1 (2')–(7') have to be entered as such because the syntactic component of the system is not powerful enough to derive them by rule from orthographic input. Moreover, in many cases semantic and pragmatic rules would be required to generate the correct prosodic output. The symbolic prosody markers trigger hierarchical sets of symbolic distinctive feature rules, followed by sets of parametric F_0 and duration rules in the phonetic-to-acoustic output component. In the case of F_0, the rules define significant points for peak and valley configurations, synchronize them with lexically stress-marked vowels according to sentence-stress and intonation symbolizations, and modify them contextually as well as microprosodically. A cosine function then interpolates between the final sequence of significant F_0 values.

The speed-control digit at the [p:] marking attributes a parametric rate variable to every segmental symbol within its domain and sets it to a value representing the respective category. Blocks of duration, segment, and F_0 rules in the phonetic module are then activated by the particular rate-variable value and the appropriate calculations along the three phonetic scales are performed. This means that for a particular speed it is not only the segment durations that are adjusted across the whole chain to which the particular rate factor applies, but F_0 is also raised for speeding up or lowered for slowing down, and segmental reductions or elaborations are effected simultaneously, in accordance with natural speech production. The segment durations are scaled separately for vowels and consonants and also as

a function of a number of other conditioning factors (vowel height, consonant category, stress, number of syllables in the word). The digit before [p:] controlling phrase-final lengthening triggers a more local increase or decrease of segment durations within the set global speech rate.

The TTS implementation of KIM allows the calculation of speech timing at a hierarchy of levels from segment to segment chain to phrase to utterance, according to a Klatt type model for segment timing with factors determined by stress, utterance position, number of syllables in the word, and overall speech rate (see [Kla79, Koh86a, Koh88]).

The Kiel prosodic model for German is comprehensive and detailed enough for its TTS realization to be capable of generating highly intelligible and natural-sounding synthetic output for very intricate phrasing structures in complicated continuous text.

37.3 A Development System for Prosodic Modeling, Prosodic Labeling and Synthesis

The TTS implementation of KIM constitutes a research tool for the further development of the prosodic model as well as for the improvement of prosodic synthesis. The categories of the prosodic phonology of German, based on extensive speech production and perception experiments, can be tested in quick interactive auditory evaluations and in formal, more costly listening experiments with carefully prepared synthetic speech output files. The development system allows rule-driven parameter control by symbolic input categories of the model as well as systematic changes of values in graphic parameter displays, in both cases with immediate acoustic output. It is thus possible to check (a) the validity of category differentiations, (b) the need for category extensions or reductions, and (c) the adequacy of defined parameter values for the categories. The results can be incorporated in a revised version of the prosodic TTS rules, and this loop of interactive or formal listening test evaluations and rule adjustments can be repeated until an optimization in the intelligibility and naturalness of the synthetic output is achieved.

By referring parametric variables to phonological categories of a prosodic model, the degrees of freedom of the TTS generation have been reduced considerably, without forgoing the flexibility and potentially exhaustive coverage of empirical speech data that a generative framework provides. However, the degrees of freedom in category combination and concatenation, such as in connection with phrase boundaries, are still quite large. The TTS development system offers a powerful device for the testing of constraints between prosodic categories. KIM will have to be preceded by a filter within the linguistic environment of the model (section 37.1.5) that excludes a great many combinations on syntactic, semantic, and pragmatic grounds and thus reduces their degrees of freedom.

The KIM prosodic symbolization system is also used for consistent, systematic, and efficient prosodic labeling of recorded speech data to enlarge the empirical

basis for prosodic modeling. For an outline of the labeling framework, its conventions, and its application see [KPS94, Koh95].

The application of the TTS research platform to prosodic model and synthesis evaluation is thus also guided by natural speech data labeled within the same category and symbolization framework. So prosodic modeling, its TTS implementation and testing, as well as model-driven labeling of natural speech data form an interrelated and mutually conditioning set of procedures in prosodic research at IPDS Kiel.

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Appendix: Audio Demos

Illustrations of the prosodic categories of KIM in the input strings to the RUL-SYS/INFOVOX TTS system: words, sentences, and texts. The words and sentences follow the numbering in section 37.2.1. All segmental and prosodic transcription notations are enclosed in # #. The accompanying CD-ROM provides the acoustic TTS output of all the listed materials; each block of word and sentence examples is presented for auditory comparison and repeated.

I. Words and Sentences

(1') Lexical stress and compounding

(a) *Rűck-sicht.* (view to the rear)
#R'YKSICHT#. (consideration)

(b) *Frei-tag.* (free day) *Freitag.* (5th day of the week)

(c) *Feier-tag.* *Donnerstag.*

(d) (wrong) reversal of the compounding in (c)

(2') Sentence stress

Max hat einen Brief #2# geschrieben.
Max hat einen Brief #1# geschrieben.
Max hat einen Brief #0# geschrieben.
Max hat einen #3# Brief #0# geschrieben.

(3') Intonation

(a) *ja. ja,. ja? ja„. ja,?*

(b) *Wie heißt du?*
Wie #2# heißt du?
Wie #2)# heißt du?

(c) *#2# Wie heißt du?* *#2)# Wie heißt du?*

(d) *Wie heißt du,.* *Wie #(# heißt du,.*

(4') Peak and valley positions

(a) peaks
Sie hat ja #)# gelogen.
Sie hat ja gelogen.
Sie hat ja #(# gelogen.

(b) high valleys in (3') (b), (c)

(c) low valleys in (3') (d)

(5') Prosodic phrase boundaries

(a) *zehn #211p:# minus #+# zwei #100p:# mal #+# drei.*
zehn #100p:# minus #+# zwei #211p:# mal #+# drei.

(b) *zehn #110p:# minus #+# zwei #000p:# mal #+# drei.*
zehn #000p:# minus #+# zwei #110p:# mal #+# drei.

(c) contrasting

i. absence of phrase boundary
ii. phrase boundary marked by lengthening
iii. phrase boundary marked by resetting of downstep
iv. phrase boundary marked by lengthening and resetting of downstep

rote gelbe blaue weiße graue schwarze.
rote gelbe #110p:# blaue weiße #110p:# graue schwarze.
rote gelbe #+100p:# blaue weiße #+100p:# graue schwarze.
rote gelbe #+110p:# blaue weiße #+110p:# graue schwarze.

(6') Speech rate: slow, medium, medium reduced, fast

mit roten gelben blauen braunen #3212p:#.
mit roten gelben blauen braunen #2212p:#.
mit roten gelben blauen braunen #1212p:#.
mit roten gelben blauen braunen #0212p:#.

(7') Downstep (default) and reset

mit roten gelben #110p:# blauen schwarzen #212p:#.
mit roten gelben #+110p:# blauen schwarzen #212p:#.

II. Texts

1. *"Die Buttergeschichte."* Medium speed, not reduced.

Es war in Berlin zu einer Zeit, als Lebensmittel nicht genügend vorhanden waren. Vor einem Laden stand bereits um sieben Uhr eine beachtliche Menschenmenge; denn man hatte dort am Abend vorher auf einem Schild schon

lesen kőnnen, daß frische Butter eingetroffen sei. Jeder wußte, daß die Butter schnell ausverkauft sein würde, und daß man ganz früh kommen műsse, um noch etwas zu erhalten. Da das Geschäft erst um acht geőffnet wurde, stellten sich die Leute vor der Laden-tűr in einer Reihe an. Wer später kam, mußte sich hinten anschließen. Je näher der Zeiger auf acht kam, desto unruhiger wurden die #)# Leute. Da kam endlich ein kleiner Mann mit grauem Haar und drängte sich ziemlich rücksichtslos nach vorn. Die wartenden Menschen waren empőrt über solches Verhalten und forderten ihn auf, sich ebenfalls hinten anzustellen. Aber auch als mit der Polizei schon gedroht wurde, #0# ließ sich der Mann nicht beirren, sondern drängte sich weiter durch. Er bat, man solle ihn doch #3# durchlassen. Oder glaubte man, daß diese Drängelei für ihn vielleicht ein Vergnügen sei?..Das war für die Leute nun doch zu viel.. Alle kochten bereits vor Wut, und der Mann konnte jetzt von allen Seiten Schimpfwőrter hören.. Er aber zuckte resigniert mit den Schultern und bemerkte : "Nun #3)# gut. Wie Sie #)# wollen. Wenn Sie mich nicht vorlassen, dann kann ich die Tűr nicht aufschließen, und Sie kőnnen meinetwegen hier stehenbleiben, bis die Butter ranzig geworden ist."

2. Address to a tutorial at the Konvens meeting in Vienna, 27 September 1994. In this text <;;> is shorthand for #000p:# (the hat pattern).

   Meine sehr ;; verehrten ;; Damen und Herren, #1# liebe ;; Teilnehmer am Tutorium #+211p:#, Aussprache-lexika in der signalnahen Sprachverarbeitung.. Es #2#begrüßt Sie die klare #+211p:#, etwas metallische #+210p:#, aber dennoch melodische #+211p:#, und vor allen ;; Dingen ;; rhythmische #+3210p:# synthetische Stimme des Nordens. Sie #2# basiert auf dem TTS-System #+2100p:# VOX #1# PC #+211p:#, der #0# Firma Infovox in Stockholm #+2212p:# und auf dem Softwehr Entwicklungswerkzeug Ruhlsys #+210p:#, der Technischen #3# Hochschule #0# Stockholm. Entscheidend für die Geburt dieser Stimme war aber die Entwicklung von Regeln #+210p:#, zur akustischen Wandlung orthographischer Symbolketten #+211p:#, im Institut für Phonetik #+210p:#, und digitale Sprachverarbeitung #+211p:#, der Christian ;; #'ALBR{CHTS%UNIVERZIT'[:T# zu Kiel...
   An #2# sich sollte #0# Professer #3# Kohler diese #1# einleitenden #1# Worte #0# sprechen. Da er aber noch ein #0# bißchen unter Zeitverschiebung #+2210p:# nach einer USA und einer Japan-reise #0# leidet, #0# braucht er #1# heute morgen #+210p:#, noch einen etwas längeren Anlauf. #0# Deshalb ist er sehr froh #0DAR'Y:Br#„ daß er diese Aufgabe #2# mir #+2210p:# einer Sprechmaschine übertragen kann....
   Ehe Herr Kohler #+2100p:# mit Ihnen die Struktur #2110# und die Generierung #2110p:# von Aussprache-lexika #+212p:#, sowie ihren Einsatz in Forschung und Anwendung #+2000p:# auf verschiedenen ;; Ebenen erläutert #+212p:#, möchte ich nicht versäumen, den Organisatoren der #1# Tagung #211p:#, auch in #1(# seinem #010p:#, Namen #+011p:#, für die Einladung zur Ausrichtung des Tutoriums #+2100p:# sehr herzlich zu ;; #)#

danken.. #2# Ihnen, den Teilnehmern, gebűhrt ebenfalls #0# Dank #210p:#, daß Sie sich #0# dafűr entschieden haben..
Wir haben die Einladung #0#natűrlich sehr ;; gern #1#aufgegriffen #+2210p:# #2# nicht nur weil sie Herrn Kohler die Mőglichkeit #1# gibt #200p:#, #3# Wien zu #1#besuchen #+211p:#, und den Heurigen zu #1# genießen #+211p:#, sondern um vor allem die Kieler ;; Forschung #1# vorzustellen #+2210p:# und #)# Interesse an ihr zu #1# wecken....
Jetzt darf ich aber Herrn Kohler nicht lűnger von seiner Arbeit abhalten. Er ist #0# inzwischen aufgewacht #2111p:# und schon #3(# unruhig geworden. Und er findet vor allem die Text-eingabe sehr erműdend #+2211p:# da er nur mit zwei Fingern tippen kann. Ich ziehe mich also zurűck, und wűnsche Ihnen viel Vergnűgen ;; beim Tutorium.